

OSSP-WEEK2-PRACTICAL

Task1: Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Source code:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    printf("Source file opened successfully.\n");
    printf("Destination file opened successfully.\n");

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            if (bytesWritten != bytesRead) {
                perror("Error writing to destination file");
                close(source);
                close(destination);
                return 1;
            }
        }

        if (bytesRead < 0) {
            perror("Error reading source file");
        } else {
            printf("File contents copied successfully.\n");
        }

        close(source);
        close(destination);

        printf("Files closed successfully.\n");

        return 0;
    }
}
```

Output:

